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EAR VISUALIZATION AND/OR TREATMENT

Abstract

A device for visualizing a surgical procedure in an ear may include a handle, a visualization shaft extending from the handle and including a distal end disposed at an angle that is not perpendicular to a longitudinal axis of the visualization shaft, and a rotation mechanism connecting the visualization shaft to the handle. The rotation mechanism allows the visualization shaft to rotate around the longitudinal axis relative to the handle. The device also includes an imaging sensor at a distal end of the visualization shaft and a light source at the distal end of the visualization shaft. The imaging sensor and the light source are positioned at the angle of the distal end relative to the longitudinal axis of the visualization shaft.

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Background/Summary

BACKGROUND

[0001] Middle ear surgery is performed on patients for a number of different reasons, most commonly for chronic recurring ear infections. When performing middle ear surgery, the ear, nose, and throat (ENT) surgeon (or “otolaryngologist”) typically visualizes the middle ear and the surgical procedure in one of two ways. In some cases, the surgeon uses a microscope, positioned in front of the surgeon's eyes, and she typically uses her non-dominant hand to hold a suction device and her dominant hand to hold a surgical tool. In other cases, the surgeon uses a handheld endoscope to visualize the middle ear. The use of a handheld endoscope is problematic in several different ways.

[0002] First, standard endoscopes have long shafts and are not made for use in the ear. When the ENT surgeon uses an endoscope in the ear, he has to hold the handle of the scope up in the air, over the patient's head, with the surgeon's hand suspended in the air, unsupported. This factor alone is problematic, because if the surgeon accidentally moves his unsupported hand during surgery, he could very easily move the distal end of the endoscope in a way that could damage the tympanic membrane or one or more of the delicate structures of the middle ear.

[0003] Second, compounding on the first issue, endoscopes are generally much heavier than the small, thin surgical tools used in middle ear procedures. The surgeon thus has an ergonomic imbalance between a relatively heavy endoscope in her non-dominant hand and a relatively light surgical device in her dominant hand. This imbalance adds to the difficulty in stabilizing the endoscope. Additionally, holding a heavy endoscope suspended over the patient's head during a surgical procedure may quickly lead to arm and hand fatigue for the surgeon.

[0004] Third, since the surgeon is holding the endoscope in one hand, that hand is no longer free to hold a suction device or a surgical tool. Thus, when an endoscope is used for visualization, the surgeon cannot use a suction device and visualize the inside of the ear at the same time.

[0005] Fourth, standard endoscopes have straight shafts, so the surgeon must hold the endoscope in a direct line straight back from the patient's ear. This straight-line position makes it impossible, or at least incredibly challenging, to use an endoscope and a microscope in the same surgical procedure, since the position of the endoscope is directly in the path of vision of the microscope. This is a drawback, because in some procedures an ENT surgeon would like to be able to switch back and forth quickly and easily between viewing with a microscope and viewing with an endoscope. It is also challenging for a surgeon to manipulate multiple tools with straight shafts held in two hands during an ear surgery procedure, because the hands must be held very close together (due to the small diameter of the ear canal), and the tools tend to bump into one another as the surgeon manipulates them to perform the procedure.

[0006] For at least these reasons, it would be advantageous to have an improved system and method for ear visualization. Ideally such a system and method would be easy to use, allow for good visualization of the ear, and be compatible with use of other ear surgery devices. At least some of these objectives will be addressed in this disclosure.

BRIEF SUMMARY

[0007] This disclosure describes various embodiments of an ear endoscope device, system, and method for visualizing an ear surgery procedure. In some embodiments, the ear endoscope device includes a handle, a main visualization shaft that holds a visualization component (e.g., a camera), and a tool attachment mechanism for coupling a surgical tool with the ear endoscope. Some embodiments also include the surgical tool itself. Specifically, several embodiments include a visualization component and a suction component, which work together as one device. The suction component (or other tool in alternative embodiments) may be coupled with the visualization

component in a number of different ways, such as by a sheath, via one or more tubes or lumens, via one or more apertures in the handle, through the main visualization shaft, etc.

[0008] The device is configured to be held in, and operated with, one hand, and it is short enough and thin enough to be advanced easily into the ear canal and to allow the surgeon to rest her hand on the patient's head during the ear procedure while holding the device, which surgeons often do with their tool-holding hand in ear surgery procedures for stability. In various embodiments, the camera, or in other words the visualization shaft, may be free to roll (or “spin”) about its own axis within a sheath or within/relative to the handle. Alternatively or additionally, the visualization shaft may be free to rotate around the longitudinal axis of the suction device within the sheath or handle, and/or the attached suction device or other surgical tool may be free to spin around the visualization shaft and/or around its own axis.

[0009] In one aspect of the disclosure, a device for visualizing and providing suction for a surgical procedure in an ear may include a handle, a visualization shaft extending from the handle and defining a longitudinal axis, an imaging sensor at a distal end of the visualization shaft, a light source at the distal end of the visualization shaft, a suction shaft extending from the handle parallel to the longitudinal axis of the visualization shaft, and a spring coupled with the suction shaft and/or the handle, such that when the suction shaft is advanced in a distal direction and then released, the suction shaft retracts automatically. Some embodiments may further include a thumb depress member coupled with a proximal end of the suction shaft, where the spring is disposed over a proximal portion of the suction shaft, between a top of the handle and a bottom of the thumb depress member.

[0010] In some embodiments, the suction shaft includes a straight proximal portion that extends parallel to the longitudinal axis of the visualization shaft and a distal curved portion, where the suction shaft is coupled with the handle such that it can spin and thus cause the distal curved portion to point in different directions. Such an embodiment may further include a thumb depress member coupled with a proximal end of the suction shaft. Optionally, a surface feature may be included on a top surface of the thumb depress member for facilitating a user spinning the thumb depress member to spin the suction shaft. In some embodiments, the device may also include a handle suction port on the handle, a free spin suction member disposed over a proximal portion of the suction shaft such that the free spin suction member does not spin when the suction shaft spins, a suction shaft suction port on the free spin suction member, and a suction tube connecting the handle suction port with the suction shaft suction port. In some embodiments, the free spin suction member houses two O-rings positioned above and below a hole in the suction shaft that communicates with the suction shaft suction port on the free spin suction member, and the O-rings and the free spin suction member form a seal with the suction shaft over the hole.

[0011] In various embodiments, the suction shaft has an outer diameter of no more than 1.1 millimeter. Some embodiments further include a suction shaft guide positioned on one side of the visualization shaft, where the suction shaft extends through the suction shaft guide. In some embodiments, the device includes a first suction shaft guide positioned on a first side of the visualization shaft, and a second suction shaft guide positioned on a second side of the visualization shaft, where the suction shaft may be passed through either the first suction shaft guide or the second suction shaft guide to provide suction on either side of the visualization shaft.

[0012] In some embodiments, the suction shaft includes a sharp distal tip for piercing a tympanic membrane, and the device further comprises a stop member on the suction shaft for preventing an ear tube positioned on the suction shaft from sliding proximally past the stop member along the suction shaft. In some embodiments, the handle is adjustable from a straight configuration to an angled configuration. Other embodiments may include a handle angle adjustment member removably attachable to the handle to adjust an angle by which the handle is held by a user.

[0013] In some embodiments, the suction shaft extends alongside the visualization shaft. Alternatively, the suction shaft may extend through the visualization shaft. The handle may include

at least one suction tube port for attaching a suction tube between the handle and the suction shaft. The handle may also include at least one suction finger control port, configured to allow a user to control application of suction by placing a finger over the finger control port and releasing the finger from the finger control port. Additionally, the handle may include a finger loop configured to allow the device to be held by a single finger of the user. In one embodiment, the finger loop is configured to extend around a middle finger of one hand of the user, the device further includes a thumb depress member to be manipulated by the thumb of the same hand, and the handle further includes a suction control port configured to be covered by the index finger of the same hand.

[0014] In another aspect of the disclosure, a device for visualizing a surgical procedure on an ear may include a suction tube, a camera coupled with the suction tube in a side-by-side arrangement, and a sheath disposed around an outside of the suction tube and an outside of the camera to couple the suction tube and the camera together. In some embodiments, the sheath holds the camera and the suction tube in such a way that the camera is free to roll or spin about its own axis within the sheath, and the camera is also free to rotate about a longitudinal axis of the suction tube within the sheath. In some embodiments, the suction tube may have an outer diameter of no more than about 1.1 millimeters, and the camera may have an outer diameter of no more than about 2.5 millimeters. In some embodiments, for example, the sheath is disposed around the camera and the suction tube but is not fixedly attached to either one, so they are free to roll and rotate within the sheath. For example, the surgeon may want to roll the camera for image orientation and/or may want to rotate the camera around the suction tube for ergonomic reasons, such when moving the device from one hand to the other.

[0015] In some embodiments, the suction tube is rigid and includes a tubular portion with a distal suction end, a suction device attachment end opposite the distal suction end, and a bend in the tubular portion. In some embodiments, for example, the tubular portion is located about 40-100 millimeters from the distal suction end. In one embodiment, the bend in the tubular portion forms an angle of about 45 degrees, although other angles are possible in alternative embodiments. In some embodiments, the sheath is shorter than a distance from the distal suction end to the bend in the tubular portion, and the camera and the sheath are configured to slide along the tubular portion of the suction tube from a first position, in which a distal end of the camera is adjacent to the distal suction end of the tubular portion, and a second position, in which the distal end of the camera is proximal to the distal suction end. In some embodiments the suction tube is made of metal. In some embodiments, at least a portion of the camera may be flexible. In some embodiments, the sheath is made of a heat-shrink polymer.

[0016] In another aspect of the disclosure, a method for performing a surgical procedure on an ear of a patient involves holding a combined visualization and suction device in one hand and advancing a distal end of the combined visualization and suction device into the ear. The combined visualization and suction device may be the same as or similar to the one described immediately above, and it may have any or all of the features described above. The method also involves viewing using the camera to view inside of the ear, activating the suction tube inside of the ear, and performing the surgical procedure on the ear, using a surgical tool held in the hand that is not holding the combined visualization and suction device. The method may also involve using the activated suction tube of the device to hold and move one or more structures within the ear. The activated suction tube may alternatively or additionally be used to suction fluid from the ear.

[0017] In some embodiments, the method may further involve rolling the camera about its own longitudinal axis within the sheath. The method may also involve rotating the camera around a longitudinal axis of the suction tube within the sheath. In some embodiments, the method may involve additionally viewing the ear using a microscope. Optionally, the suction tube may include a bend, and the method may further involve holding the combined visualization and suction device outside of a direct line of sight between a surgeon's eyes and the ear. The method may also involve supporting the hand that is holding the combined visualization and suction device on the patient's

head during the surgical procedure. Optionally, the method may involve supporting the hand that is holding the surgical tool on the patient's head during the surgical procedure.

[0018] In another aspect of the present disclosure, a device for visualizing a surgical procedure in an ear may include an ear endoscope and a coupler. The ear endoscope includes a handle, a shaft extending from the handle and having a bend with an angle of 90-155 degrees, an outer diameter of no more than 2.5 millimeters, and a length of 30-80 millimeters, an imaging sensor at a distal end of the shaft, and a light source. The coupler is attached to a side of the ear endoscope shaft for attaching a tool to the endoscope. In various embodiments, the surgical tool and the overall device may include any of the features described above. The surgical tool may be a suction device, as previously described, or alternatively it may be any other suitable tool, such as but not limited to a cutting device, a piercing device, an ear tube placement device, a seeker, tweezers or forceps.

[0019] In another aspect of the disclosure, a method for performing a surgical procedure in an ear of a patient may first involve attaching a tool to an ear endoscope in a side-by-side arrangement, using a coupler, where the ear endoscope includes a shaft with a bend and an outer diameter of no more than 2.5 millimeters. The method may further involve holding a handle of the ear endoscope in one hand, advancing a distal end of the ear endoscope into the ear with the tool attached, viewing an inside of the ear, using the ear endoscope, and using the tool attached to the ear endoscope to facilitate or perform at least part of the surgical procedure. The combined visualization and surgical tool device may be the same as, or similar to, the embodiment described above, and it may include any of the features described above.

[0020] In another aspect of the present disclosure, an ear endoscope device for use in a surgical procedure in an ear may include a handle, a visualization shaft extending from the handle, a tool guide extending from the handle parallel to the visualization shaft and configured to guide a tool into the ear with the visualization shaft, an imaging sensor at a distal end of the visualization shaft, and a light source. In one embodiment, the ear endoscope device may include at least one tool coupler on a side of the shaft, at least one suction shaft insertion port at or near a distal end of the handle, two side suction tube connection ports at or near the distal end of the handle, a rear suction tube connection port at or near a proximal end of the handle, and a suction lumen connecting the rear suction tube connection port to the two side suction tube connection ports. In various embodiments, the shaft and the handle may form an angle of between about 90 degrees and about 155 degrees. In some embodiments, the shaft may have an outer diameter of no more than about 2.5 millimeters and a length of between about 30 millimeters and about 80 millimeters.

[0021] In some embodiments, the ear endoscope further includes a suction device. The suction device may include a suction shaft for passing through the at least one suction shaft insertion port and the at least one tool coupler, a thumb depress member coupled with the suction shaft for allowing a user to advance the suction shaft, a side suction tube for attaching the suction shaft, via the thumb depress portion, to one of the two side suction tube connection ports, and a rear suction tube for connecting the rear suction tube connection port to a suction source. The suction device may further include a spring disposed over a proximal portion of the suction shaft, between the thumb depress member and the handle of the ear endoscope. The spring may be configured to automatically retract the suction shaft relative to the shaft when the thumb depress portion is released. In some embodiments, an open one of the two side suction tube connection ports that is not attached to the side suction tube is configured to act as a finger operated suction control for controlling the application of suction force with a user's finger.

[0022] In some embodiments, the handle includes two suction shaft insertion ports and two tool couplers disposed on opposite sides of the shaft, where each of the two suction shaft insertion ports feeds into a corresponding one of the two tool couplers. In some embodiments, the handle includes a finger loop for facilitating holding the device with a user's finger under the handle. Alternatively, the handle may include any other finger hold shape or other ergonomic shape to facilitate gripping the device with one hand.

[0023] In another aspect of the present disclosure, a method for performing a surgical procedure in an ear canal of a patient may involve holding in one hand an ear endoscope with an attached suction device, advancing a distal end of the ear endoscope with the attached suction device into the patient's ear canal, depressing a thumb depress member of the suction device with a thumb of the hand, to advance a suction shaft of the suction device relative to a visualization shaft of the ear endoscope, applying suction in the ear canal with the suction device, and viewing an inside of the ear canal, using the ear endoscope. In one embodiment, applying suction in the ear canal involves applying a finger of the hand to an open suction control opening on the handle.

[0024] In some embodiments, the method also involves releasing the thumb depress portion to allow a spring on the suction shaft to expand to cause the suction shaft to retract relative to the shaft of the ear endoscope. In some embodiments, depressing the thumb depress member causes the suction shaft to advance through a suction shaft insertion port on a handle of the ear endoscope and through a tool coupler attached to the shaft of the ear endoscope. The spring may be disposed over the suction shaft, between the thumb depress member and the handle. The method may optionally also involve supporting the hand that is holding the ear endoscope on the patient's head during the surgical procedure.

[0025] These and other aspects and embodiments are described in further detail below, in relation to the attached drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a perspective view of a surgeon's hands and a surgical field, including a patient's ear, illustrating how a prior art endoscope and surgical tool are typically held;

[0027] FIG. 2 is a side view of an ear visualization system, shown with a suction device, according to one embodiment;

[0028] FIG. 3 is a close-up illustration of a distal portion of the ear endoscope shown in FIG. 2;

[0029] FIG. 4 is a perspective view of coupler for use with an ear visualization system, according to one embodiment;

[0030] FIG. 5A is a side view of a combined visualization and suction device for use in ear surgery procedures, according to one embodiment;

[0031] FIG. 5B is a side view of the visualization/suction device of FIG. 5A, with the suction component advanced distally, relative to the visualization component;

[0032] FIG. 5C is a side, exploded view of the visualization/suction device of FIGS. 5A and 5B;

[0033] FIG. 5D is a front, end-on view of the shafts of the combined visualization and suction device of FIGS. 5A-5C;

[0034] FIG. 6 is a perspective view of a surgeon's hands and a surgical field, including a patient's ear, illustrating how a combined visualization and suction device, according to one embodiment, may be held and used during an ear surgery procedure; and

[0035] FIG. 7 is a side view of a combined visualization and suction device for use in ear surgery procedures, where the camera has a bend near its distal end, according to one embodiment;

[0036] FIG. 8 is a side view of a combined visualization and suction device for use in ear surgery procedures, where the sheath couples the camera and suction tube together at an angle relative to one another, according to another embodiment;

[0037] FIGS. 9A-9C are side, front and exploded views, respectively, of a combination visualization and suction device, according to an alternative embodiment;

[0038] FIGS. 10A and 10B are side views of a portion of the combination device of FIGS. 9A-9C, illustrating a method for advancing and retracting a suction shaft relative to a visualization shaft, according to one embodiment;

[0039] FIG. **11** is a perspective view of portion of a patient's head, including the ear, and a physician's hand holding the combination device of FIGS. **9A-10B**;

[0040] FIG. **12** is a perspective view of a visualization component of the combination device of FIGS. **9A-11**, along with a viewing system, according to one embodiment;

[0041] FIGS. **13A-13C** are right/rear perspective, left/rear perspective and left side views, respectively, of an ear endoscope device, according to one embodiment;

[0042] FIG. **14A** is a side view of the ear endoscope device of FIGS. **13A-13C**, with a detached optional suction device for the ear endoscope, according to one embodiment;

[0043] FIG. **14B** is a side view of the ear endoscope and suction device of FIG. **14A**, with the suction device attached to the ear endoscope;

[0044] FIG. **14C** is a perspective view of the ear endoscope with suction device of FIGS. **14A** and **14B**, shown in the left hand of a physician user;

[0045] FIG. **15** is a distal-end view of a shaft of an ear endoscope, according to an alternative embodiment;

[0046] FIGS. **16A** and **16B** are side views of an ear endoscope with a curved tip suction shaft, showing the suction shaft advanced distally (FIG. **16A**) and retracted proximally (FIG. **16B**), according to one embodiment;

[0047] FIG. **17** is a side view of an ear endoscope with a curved tip suction shaft and a handle with a finger loop, according to an alternative embodiment;

[0048] FIGS. **18A-18D** are perspective views of four different embodiments of a thumb depress member of an ear endoscope with a curved tip suction shaft;

[0049] FIG. **19** is a side view of an ear endoscope device with an adjustable-angle handle, according to one embodiment;

[0050] FIG. **20** is a side view of an ear endoscope device with a malleable handle, according to one embodiment;

[0051] FIG. **21A** is a side view of an ear tube placement, visualization and suction device, advanced partway into an ear canal, according to one embodiment;

[0052] FIG. **21B** is a side view of the device of FIG. **21A**, showing placement of an ear tube across the tympanic membrane, according to one embodiment;

[0053] FIG. **22** is a side view of a curved distal end of an ear tube placement, visualization and suction device, according to one embodiment; and

[0054] FIGS. **23** and **24** are side views of a portion of an ear visualization/suction device, illustrating a removable handle adjustment feature (FIG. **24**);

[0055] FIGS. **25** and **26** show the portion of the device from FIGS. **23** and **24** without (FIG. **25**) and with (FIG. **26**) an optional handle adjustment feature;

[0056] FIGS. **27A** and **27B** are diagrammatic illustrations of an ear visualization/suction device of the present application, shown in an orientation as if the device were being held in a left hand while inserted into a right ear (FIG. **27A**) and a left ear (FIG. **27B**), according to one embodiment;

[0057] FIGS. **28A** and **28B** are diagrammatic illustrations of an ear visualization/suction device of the present application, shown in an orientation as if the device were being held in a left hand while inserted into a right ear (FIG. **28A**) and a left ear (FIG. **28B**), according to an alternative embodiment;

[0058] FIGS. **29A** and **29B** are front perspective and side perspective views, respectively, of an ear visualization and suction device, according to an alternative embodiment;

[0059] FIG. **29C** is a side view of a suction shaft portion of the device of FIGS. **29A** and **29B**;

[0060] FIG. **30** is a side view of two different suction shaft portions of the device of FIGS. **29A** and **29B**;

[0061] FIG. **31A** is a rear perspective view of the device of FIGS. **29A** and **29B**, with an alternative suction shaft portion attached and with part of the handle removed;

[0062] FIGS. **31B** and **31C** are top views of a portion of the configuration of the device of FIG.

31A;

[0063] FIG. 32A is a side view of an ear visualization and suction device with optional irrigation, according to one embodiment;

[0064] FIG. 32B is a side view of the device of FIG. 32A, with the suction component not shown and the irrigation portion removed from the handle;

[0065] FIG. 32C is a side view of the device of FIGS. 32A and 32B, with the suction component not shown and the irrigation portion inserted into the handle;

[0066] FIG. 32D is a top view of the device of FIGS. 32A-32C, in the configuration of FIG. 32C;

[0067] FIG. 33 is a perspective view of an ear endoscope device with a rotating shaft and angled distal end and camera, according to one embodiment;

[0068] FIG. 34 is a close-up perspective view of a distal portion of the shaft of the ear endoscope device of FIG. 33;

[0069] FIG. 35 is a close-up perspective view of a rotation mechanism attached to a proximal end of the shaft of the ear endoscope device of FIGS. 33 and 34;

[0070] FIG. 36 is a partial phantom view of the rotation mechanism of FIG. 35; and

[0071] FIG. 37 is a perspective view of an ear endoscope device with a rotating shaft and angled distal end and camera, according to an alternative embodiment.

DETAILED DESCRIPTION

[0072] In general, the embodiments described herein are directed to a device, system, and method for visualizing an ear surgery procedure. The ear visualization device generally includes an ear endoscope (or “camera”), housed in a visualization shaft, with an attachment mechanism for attaching an additional tool to the visualization shaft. Oftentimes, the additional tool is a suction device, so the device provides for visualization and suction with one device, held in one hand. In alternative embodiments, however, any of a number of different tools may be attached to the endoscope/visualization shaft, in addition to or instead of a suction device. In some embodiments, the attachment mechanism for attaching the additional tool is built into the endoscope. For example, the attachment mechanism is the handle in some embodiments. Alternatively, the attachment mechanism may be a separate coupler or sheath, which attaches to the visualization shaft of the ear endoscope device and allows any of a number of different types of surgical tools to be attached to the endoscope/visualization shaft in a side-by-side arrangement. In yet other embodiments, visualization and suction may be integrated into the device, for example via the handle. The ear visualization system may include the ear endoscope along with a separate attachment mechanism, a light source for the endoscope, a video monitor for displaying images captured by the endoscope and/or any other suitable components. In some embodiments, the system may also include a suction device or other surgical tool. In other embodiments, the ear endoscope device or system may be provided by itself, and may be used with one or more optional, stand-alone tools.

[0073] As mentioned immediately above, in some embodiments, the attachment mechanism (or “coupler”) is a separate piece, which may be removed from the endoscope/visualization shaft. In such embodiments, the endoscope and the coupler may be referred to as a “system,” due to the combination of two different devices. In alternative embodiments, the coupler may be integral with, or permanently attached to, the endoscope/visualization shaft, in which case the endoscope with coupler may be referred to as a “device.” In any case, use of the terms “system” and “device” herein should not be interpreted as limiting the scope of the invention.

[0074] In some embodiments, the device may include an endoscope and a suction tube that also operates as an ear tube placement device. The distal tip of the suction/ear tube placement component may have a sharp distal tip to pierce the tympanic membrane, and the device may also include a stop for preventing the ear tube from sliding proximally up the suction/ear tube placement component. This device embodiment is described in further detail below.

[0075] The visualization shaft of the ear endoscope device and whatever surgical tool it is used

with may have very small diameters, so the distal end of the device fits easily into an ear canal, for helping visualize and perform an ear surgery procedure. In some embodiments, the coupler surrounds part of the visualization shaft and part of the surgical tool in such a way that the shaft can rotate about a longitudinal axis of the tool and can also roll (or “spin”) about its own longitudinal axis.

[0076] In one embodiment, described in detail below, the surgical tool is a suction tube device. In alternative embodiments, however, the tool may be any suitable, small-diameter tool, such as but not limited to an irrigation tube or shaft, a cutting device, a piercing device, an ear tube placement device, a seeker, tweezers, forceps, a speculum, a grasper, or a curette. In the description below of the suction embodiment, the fact that any other suitably sized surgical tool may be substituted for the suction device will not be repeated with the description of every embodiment. Similarly, the devices and methods described below for use in an ear surgery procedure may be used or adapted for use in any other suitable surgical procedure, such as but not limited to a neurosurgery procedure, a spine surgery procedure, an orthopedic surgery procedure on a joint or other structure, and/or the like. This, too, will not be repeated with the description of every embodiment. Therefore, although the following description is focused on use of the devices, systems, and methods for visualizing and facilitating ear surgery procedures, the same embodiments may be used, or adapted for use, in any other suitable procedures and parts of a human or animal body. Therefore, the invention is not limited to use in the ear.

[0077] Referring now to FIG. 1, a prior art method for performing an ear surgery procedure using a standard endoscope **10** is illustrated. The figure shows a surgical field, with the patient's ear **E** exposed for the procedure. The surgeon is holding the endoscope **10** in his left hand **L** and a surgical tool **12** in his right hand. Due to the length of the endoscope **10**, the surgeon has to hold his left hand **L** up in the air, suspended over the patient, in order to hold the handle of the endoscope **10**. As mentioned previously, this can be very awkward and potentially dangerous to the tympanic membrane and/or structures of the middle ear, especially in longer procedures where the surgeon's left arm and left hand **L** get fatigued. Additionally, the surgeon does not have a free hand to hold a suction device or other surgical tool, since both of the surgeon's hands are occupied. To have suction in this scenario, a nurse or other assistant would have to hold the suction device in the patient's ear.

[0078] Referring to FIG. 2, an ear surgery visualization system **100**, according to one embodiment, may include an ear endoscope **102** and a coupler **112**. Also pictured in FIG. 2 is a suction device **128**, which is not necessarily part of the system **100**, but which is shown in the figure for illustrative purposes. In alternative embodiments, the suction device **128** may be replaced by any other suitable surgical tool, such as but not limited to an irrigation tube or shaft, a cutting device, a piercing device, an ear tube placement device, a seeker, tweezers, forceps, a speculum, a grasper, or a curette.

[0079] The ear endoscope **102** includes a handle **104**, a shaft **106** (or “visualization shaft”) and a processor **122**, which may also act as a light source. The shaft **106** includes a proximal portion **107**, a bend **108** and a distal portion **110**, ending in a distal tip **111**. The endoscope **102** also includes a light source **120** in the handle **104**, and light fibers **118** that carry the light from the light source **120**, through the shaft **106**, to the distal tip **111**. A camera on a chip (described more fully below) may be positioned at the distal tip **111**, to acquire images of the ear. The system **100** may also include a video monitor **126**, although optionally the video monitor **126** may be a separate component that is not part of the system **100**. In another embodiment, the processor **122** and video monitor **126** may be combined in one unit. In another embodiment, the processor **122** may be housed within the handle **104**.

[0080] The shaft **106** of the endoscope **102** may have a total length of about 30 millimeters to about 80 millimeters and an outer diameter of less than about 2.5 millimeters or less than about 5 millimeters. In some embodiments, the outer diameter of the shaft **106** may be continuous along its

length. Alternatively, the outer diameter of the distal portion **110** may be smaller than the outer diameter of the proximal portion **107**. The bend **108** may form an angle between the proximal portion **107** and the distal portion **110** of between about 90 degrees and about 155 degrees. The handle **104** may be very small and lightweight, compared to typical endoscope handles. In fact, the handle **104** may be shaped to have a comfortable pencil grip, so the surgeon may hold and manipulate the ear endoscope **102** like a pencil. Ear endoscope **102** may also include a cable **124**, attaching the handle **104** to the processor **122**.

[0081] The coupler **112** includes an endoscope attachment portion **114** and a tool attachment portion **116**. In some embodiments, each of the two portions **114**, **116** is shaped as a tube or a semicircular tube. In some embodiments, the endoscope attachment portion **114** and the tool attachment portion **116** may have the same diameter. Alternatively, they may have different diameters. For example, in some embodiments the endoscope attachment portion **114** has a larger diameter than that of the tool attachment portion **116**. The coupler **112** may be permanently attached to the shaft **106**, or it may be removable, according to different alternative embodiments. The coupler **112** may be attached to the distal portion **110** of the shaft **106**, as shown. Alternatively, the coupler **112** may be attached to the proximal portion **107**, for example if the shaft **106** is straight, or of the coupler **112** follows the bend **108** in the shaft **106**.

[0082] The weight, size and feel of the ear endoscope **102** may be similar to that of other ear surgery tools. This makes it more comfortable for the surgeon to hold and prevents an imbalance between the ear endoscope **102** and other tools. The surgeon may hold the handle **104** with a pencil grip and may rest her hand and/or the handle **104** on the patient's head during the procedure. To achieve this desired weight, size, and feel, any suitable materials may be used for the various parts of the ear visualization system **100**. For example, in one embodiment, the handle **104** may be made of any suitable lightweight plastic, and the shaft **106** may be made of any suitable metal, such as stainless steel. Alternatively, the handle **104** may be made of a lightweight metal. The coupler **112** may be made of plastic or metal, for example. Any suitable, medically safe materials may be used.

[0083] As mentioned above, the suction device **128** is illustrated in FIG. 2, attached to the shaft **106** of the endoscope **102** via the coupler **112**, in a side-by-side arrangement. Any other tool may be substituted for the suction device **128**, in alternative embodiments. The suction device **128** is also shown with a source of suction **130**, which may be a separate component, wall suction, or any suitable suction source. The suction tube portion of the suction device **128** is flexible, at least along part of its length, and has a distal portion with an outer diameter that fits within the tool attachment portion **116** of the coupler. Various embodiments and features of a suction device **128** are described in further detail below.

[0084] Referring now to FIG. 3, the distal portion **110** of the endoscope/visualization shaft **106** is illustrated in greater detail. At the distal end **111** of the shaft **106** are positioned an imaging sensor **140** and two light sources **142**. The imaging sensor **140** may be any type of suitable sensor, such as a complementary metal-oxide semiconductor (CMOS) camera or any other “camera on a chip” type of device. The two light sources **142** (or alternatively any other number of light sources) may be light emitting diode (LED) lights, for example. These may be in addition to, or as an alternative to, the light source **120** shown in the handle **104** in FIG. 2. In other words, according to various embodiments, one or more light sources for the endoscope device **102** may be located in the handle **104**, at the distal end **111** of the shaft **106**, or both.

[0085] FIG. 4 is a magnified view of an alternative embodiment of a coupler **212** for use with the ear endoscope **102**. The coupler **212** includes an endoscope attachment portion **214** for attaching to the shaft **106**, a tool attachment portion **216**, a longitudinal top opening **250** in the tool attachment portion **216**, and a longitudinal middle opening **252** between the endoscope attachment portion **214** and the tool attachment portion **216**. In this embodiment the endoscope attachment portion **214** has a larger diameter than that of the tool attachment portion **216**. The tool (not shown) may be inserted into the tool attachment portion **216** by pushing it down through the top opening **250** or by sliding

it into the proximal end of the tool attachment portion **216** and advancing it distally. In the case where the tool is pushed through the top opening **250**, the coupler **212** may flex outward slightly, by expanding at the two openings **250**, **252**, to accommodate the tool. In alternative embodiments, the endoscope attachment portion **214** and/or the tool attachment portion **216** may be formed as complete tubes, with circular cross-sections rather than semi-circular cross-sections. As mentioned above, the coupler may be made of any suitable material.

[0086] Referring to FIGS. 5A-5D, one embodiment of an ear surgery visualization device **20** is illustrated. In this embodiment, the visualization device **20**, which may also be called “a combined visualization and suction device,” includes a suction tube **22**, a camera **30** (also referred to herein as a “visualization shaft,” which includes a camera in it) and a coupler **38** (or “sheath”) disposed around the suction tube **22** and the camera **30**. The suction tube **22** has a distal end **24**, a proximal end **26** for connecting with suction tubing connected to a suction source, and a bend **28** along its length. The camera **30** includes a distal portion **32**, a proximal portion **36** and a distal end **34**.

[0087] The suction tube **22** may be any standard or customized suction tube device. In various embodiments, the suction tube **22** may be rigid and may be made of any suitable material, such as stainless steel or other biocompatible metal or plastic. The suction tube **22** will have an overall diameter and length to allow it to be advanced easily into the ear and to allow a surgeon to hold the visualization device **20** with one hand, resting on the patient's head, during the procedure. In some embodiments, for example, the suction tube **22** has an outer diameter, at least along the portion between the bend **28** and the distal end **24**, of about 0.6 millimeter to about 1.1 millimeters. The bend **28** in the suction tube **22** is optional, and alternative embodiments may be straight. The bend **28** may be advantageous, however, because it allows the visualization device **20** to be held at an angle from the ear, so the hand holding the device **20** is not in the direct line of sight of the surgeon. This is especially advantageous in cases where the surgeon wants to use a microscope and the visualization device **20** in the same procedure, but it is also advantageous in keeping the suction tube **22** and the camera **30** out of the way of any surgical tools held in the surgeon's other hand. In various embodiments, for example, the bend **28** may be located about 40 millimeters to about 100 millimeters from the distal end **24** of the suction tube **22**. In one embodiment, the bend may be about 60 millimeters from the distal end **24**. In alternative embodiments of the device **20**, where the camera **30** is combined with a different type of surgical tool rather than the suction tube **22**, that surgical tool may also include the same or a similar bend.

[0088] The camera **30** may be any suitable, small-diameter camera for viewing an ear during an ear surgery procedure. In some embodiments, for example, the camera **30** may be a fiber optic camera or a complementary metal-oxide-semiconductor (CMOS) camera. As small-diameter cameras are well known, they will not be described in detail here. In some embodiments, at least the distal portion **32** of the camera **30** may be relatively rigid, so that the surgeon can easily roll it about its longitudinal axis and/or rotate it relative to the suction tube **22**. In some embodiments, the camera **30** may include a bend, which may coincide with the bend **28** in the suction tube. The camera **30** may include CMOS sensors with a lens array. The sensors may be arrayed in a cube of between 0.6 mm by 0.6 mm and 1.0 mm by 1.0 mm, with overall length of up to 3 mm, in some examples. Alternative embodiments may include a fiber optic bundle for image capture, rather than CMOS. The light source for illumination may be LED at the distal tip **34** or fiber infused with light from a remote LED.

[0089] The cross-sectional shape of the camera **30** may vary in different embodiments (round, oval, square, rectangular, etc.), but in the embodiment shown the camera **30** has a round cross-sectional shape. This is advantageous for rolling and rotating the camera **30** within the coupler **38** and relative to the suction tube **22**. The body of the camera **30** is made from a relatively rigid or at least semi-rigid material, such as stainless steel or plastic (e.g., thermoplastic). The length of the distal portion **32** may be, for example, about 5 mm to about 100 mm. In some embodiments, the distal portion **32** may be as long as the length of the suction tube **22** from its distal end **24** to the bend **28**,

which in one embodiment is about 60 mm. In various embodiments, the camera **30** and a light source may be integrated into a metal tube, over-molded with plastic, encapsulated in a polymer, or the like.

[0090] The coupler **38** may be any suitable material and have any suitable length, thickness and size, according to various embodiments. In one embodiment, the coupler **38** is formed as a tube of heat-shrink polymer wrap that surrounds distal portions of the suction tube **22** and the camera **30**. The heat-shrink polymer may be polyethylene terephthalate (PET) in some embodiments, or may alternatively be any other suitable polymer, such as but not limited to a polyolefin, a polyimide or nylon. As illustrated in FIGS. 5A and 5B, in one embodiment, the coupler **38** is disposed about the suction tube and the camera **30** such that the suction tube **22** can advance (FIG. 5B) and retract (FIG. 5A), relative to the coupler **38** and the camera **30**. For example, in some embodiments, the suction tube **22** can advance from a position where its distal end **24** is at or near the distal end **34** of the camera **30** (FIG. 5A) to a position where its distal end **24** is ahead of that of the camera **30** (FIG. 5B). In this embodiment, the camera **30** may also be able to slide forward and backward. In alternative embodiments, camera **30**, suction tube **22** or both may be fixed to the inner surface of the coupler **38**, such as by adhesive, thus reducing the amount of mobility of one or both components relative to the coupler **38**.

[0091] FIG. 5C is an exploded view of the ear surgery visualization device **20**, showing the suction tube **22**, the camera **30** and the coupler **38** separate from one another. For assembly, the coupler **38** may be wrapped or slid over the suction tube **22** and the camera **30** in some embodiments.

[0092] FIG. 5D illustrates possible directions of movement of the camera **30**, the suction tube **22** and the coupler **38**, relative to one another. In some embodiments, the coupler **38** may be positioned around but not fixedly attached to the camera **30** and the suction tube **22**, as described above. In addition to allowing the suction tube **22** and/or the camera **30** to advance longitudinally through the coupler **38**, this configuration also allows the suction tube **22** and the camera **30** to roll about their own axes within the coupler **38** (two small hollow arrows around perimeter of coupler **38**). Additionally, the camera **30** may be rotated around a longitudinal axis of the suction tube **22** (larger hollow arrow). The suction tube **22** may also be rotated around a longitudinal axis of the camera **30**. This freedom of movement-rotation and rolling-allow the surgeon to adjust the orientation of the camera **30** and/or the suction tube **22** easily and quickly, without necessarily changing the orientation of both components. Again, however, in alternative embodiments the coupler **38** may be adhered or otherwise fixedly attached to either or both of the camera **30** and the suction tube **22**.

[0093] In some embodiments, as will be described in further detail below, the distal end **34** of the shaft of the camera **30** (i.e., the “visualization shaft”) may be beveled, and the camera **30** may be angled to face out of the beveled opening in the distal end **34**. For example, the camera **30** may have an angle of between about 5 degrees and about 90 degrees, relative to the longitudinal axis of the camera. This angle may be achieved by placing the camera **30** which may be a ‘camera on a chip’ within the tip of the visualization shaft in an angle so that it is tilted, or via one or more mirrors, prisms, swing prisms, or the like. By providing a camera **30** that is angled and can rotate around its own axis, the visualization device **20** allows a physician to rotate the camera **30** to view in a range of different directions, up to and including a 360-degree range, without turning the entire visualization device **20** in his or her hand. This facilitates viewing in multiple directions without requiring awkward device manipulation.

[0094] FIG. 6 shows a surgical field, including a patient's ear E, and the left hand L and the right hand R of a surgeon, performing a procedure on the ear E. The surgeon's left hand L is holding the ear surgery visualization device **20**, as described above, which includes the suction tube **22** and the camera **30**. The suction tube **22** is attached proximally to a suction hose **40**, which in turn is attached to a source of suction (not shown). The surgeon's right hand R holds a surgical tool **12**. As indicated by the large arrows on the figure, the surgeon's hands are approaching the ear E from two

different angles, leaving a line of direct vision open from the surgeon's eyes to the patient's ear E (depicted by the middle/upper-right hollow arrow). This arrangement will allow a surgeon to visualize the surgical field using both a microscope and the camera **30**, if desired. As also illustrated in FIG. **6**, the ear visualization device **20** is sized and shaped such that the surgeon can rest her hand on the patient's head during the procedure. In performing the procedure, the surgeon may advance the device **20** into the ear E, suction out the ear E using the suction tube **22**, visualize the ear E using the camera **30**, and perform the procedure. Alternatively or additionally, the suction tube **22** may be used to hold onto and move one or more small anatomical structures of the ear, such as but not limited to the bones of the middle ear. Suction may also be used to hold different devices, such as an ear tube or ossicular prostheses. The device **20** is generally small enough that the camera **30** can be used to visualize the middle ear through a natural hole or incision in the tympanic membrane. These actions may be performed in any sequence and in any combination. In some embodiments, it may be possible for the surgeon to separate the camera **30** from the suction tube **22** during the ear surgery procedure, so they can be used separately.

[0095] In some embodiments, the ear surgery visualization device **20** may be used with another, different ear surgery visualization device (not shown). For example, the combined camera/suction tube device **20** may be held in the surgeon's non-dominant hand, and a combined camera/surgical tool device may be held in the surgeon's dominant hand. These two devices **20** may be used at the same time, thus acquiring two images of the ear. The views from the two cameras may be displayed on a single, split video screen, for example, with the right half marked 'R' and the left half marked 'L'. In all embodiments, the video screen may be separate and located above the patient's head and within the field of view of a microscope, so that the surgeon can view the surgical field through the microscope and look at the endoscopic view through the microscope as well, or simply switch from looking through the microscope to looking at the video screen. In another embodiment, it may be possible to digitally feed the endoscopic image into the microscope, so that the surgeon can view both of them through the microscope, or toggle between them by pressing a button, for example.

[0096] Referring now to FIG. **7**, an alternative embodiment of an ear surgery visualization/suction device **50** is illustrated. As mentioned above, in some embodiments, the suction tube **22** may be advanced, relative to the camera **30**, as shown in FIG. **5B**, such that the distal end **24** of the suction tube **22** is ahead of the distal end **34** of the camera **30**. In embodiments where the distal portions of the suction tube **22** and the camera **30** are both straight and are connected in parallel with one another, the suction tube **22** may interfere with the field of view of the camera **30** in this configuration. The embodiment of the device **50** shown in FIG. **7** is configured to address that issue. In this embodiment, the distal portion **32** of the camera **30** includes a bend **52**. This bend **52** orients the field of view **54** of the camera **30** at an angle, relative to the longitudinal axes of the suction tube **22** and the camera **30**, so the suction tube **22** does not interfere with or limit the field of view **54**. In various embodiments, the bend **52** may be located anywhere along the length of the camera **30**, although in many embodiments it will be located near the distal end **34**, so that it is distal to the distal end of the coupler **38**.

[0097] Referring now to FIG. **8**, another alternative embodiment of an ear surgery visualization/suction device **60** is illustrated. This embodiment illustrates an alternative way of addressing the issue of the suction tube **22** cutting off part of the field of view **54** of the camera. In this embodiment, the sheath **68** has a wider distal end **64** and a narrower proximal end **66**. Thus, the sheath **68** couples the camera **30** and the suction tube **22** together such that they are oriented at an angle relative to one another. In other words, they are not parallel with one another. As with the previous embodiment, this helps prevent the field of view **54** of the camera **30** from being limited by the suction tube **22**.

[0098] Referring now to FIGS. **9A-9C**, one embodiment of an ear visualization system **300** is illustrated. The ear visualization system **300** includes an ear endoscope **302** and an optional suction device **310**. The ear endoscope **302** includes a handle **304**, a shaft **306** extending from one end of

the handle **304**, two tool couplers **307a**, **307b** (see FIG. **9B**) on either side of the shaft **306**, and a cable **308** extending from the opposite end of the handle **304**. Imaging components pass through the handle **304**, the shaft **306** and the cable **308**, which components may be any of those described above and which are not shown in these figures. The handle **304** includes two suction shaft apertures **305a**, **305b** (not visible in these drawings, because they are on the top surface of the handle **304**), through which the shaft **316** of the suction component **310** is advanced. The shaft **316** of the suction component **310** advances through one of the two tool couplers **307a**, **307b**, after exiting the distal end of the corresponding aperture **305a**, **305b**. The handle **304** may have any of a number of suitable sizes, shapes and weights, but in this embodiment it is configured to be held easily in a pencil grip by the physician. The handle **304** may be made of lightweight plastic, in some embodiments.

[0099] The suction component **310** includes a suction tube **312**, a thumb depress portion **314**, a suction control aperture **320**, a suction shaft **316** with a distal end **317**, and a spring **318** disposed over the suction shaft **316**, between the thumb depress portion **314** and the handle **304**. The suction shaft **316** extends through the suction shaft aperture **305a** in the handle **304**, through the tool coupler **307a**, and alongside the visualization component shaft **306**. As will be described further below, the user physician may depress the thumb depress portion **314** to advance the distal end **317** of the suction shaft **316** out of the distal end of the visualization shaft **306** and thus farther into the ear. When the user releases the thumb depress portion **314**, the spring **318** automatically retracts the suction shaft **316** back along the visualization shaft **306**, through the tool coupler **307a** and the aperture **305a**. The physician may use an index finger (or other finger) to cover the suction control aperture **320** to apply suction, and she may remove the finger from the hole to remove or reduce suction at the distal end **317** of the suction shaft **316**.

[0100] FIG. **9B** is a front view of the ear visualization system **300**, which illustrates that this embodiment includes two suction tool couplers **307a**, **307b**, one on either side of the visualization component shaft **306**. This embodiment thus also includes two suction shaft apertures **305a**, **305b**, each feeding into one of the two suction tool couplers **307a**, **307b**. The two tool couplers **307a**, **307b** facilitate holding and manipulation of the device **300** by either a right hand or a left hand, and placement of the suction tube/other tool either below or above the camera sensor. The visualization component shaft **306**, the tool couplers **307a**, **307b** and the suction shaft **316**, in some embodiments, may be made of metal, such as stainless steel or other biocompatible metal. In some embodiments, the visualization component shaft **306** has an outer diameter of about 2.5 millimeters or less. Similarly, each of the tool couplers **307a**, **307b** may have an outer diameter of about 2.5 millimeters or less. The suction shaft **316** has an outer diameter sized to fit through the inner diameter of the tool couplers **307a**, **307b**. In some embodiments, the suction shaft **316** may have an outer diameter of about 1.1 millimeters or less.

[0101] FIG. **9C** is a side view of the ear visualization system **300**, with ear endoscope **302** separated from the suction device **310**. In this embodiment, the ergonomic design of the handle **304** may be important for facilitating handling of the system **300** by the physician. For example, the handle **304** includes a finger grip feature **322**, which may allow for easy gripping of the handle **304** with a middle finger (or other finger). The user's index finger may be used to control the suction control aperture **320**, and the user's thumb may be used to control the thumb depress portion **314** of the suction component **310**. In other embodiments, one of which is described below, the finger grip feature **322** may include a loop, an elastic ring or any other suitable shape.

[0102] In various alternative embodiments, one or more variations may be made to the ear endoscope device **300**. For example, in some embodiments, the couplers **307a**, **307b** may extend the entire length (or along a longer portion but not the entire length) of the endoscope visualization shaft **306**. In some embodiments, there may be only one coupler and one aperture, rather than two couplers **307a**, **307b** and two apertures **305a**, **305b**.

[0103] Referring to FIG. **15**, in yet another alternative embodiment, an ear endoscope shaft **500**

may include an outer shaft body **502**, a tool guide **504** forming a tool lumen **506**, two light sources **508a**, **508b** and an imaging sensor. The tool guide **504** and tool lumen **506**, in this embodiment, are located inside the outer shaft body **502**, unlike the previously described embodiments that place the suction tube through a coupler on the outside of the main endoscope shaft. In some embodiments, the tool guide **504** may be used for applying suction or advancing a suction device through the shaft **500**. Alternatively, the tool guide **504** may be used for advancing any other suitable tool through the ear endoscope shaft **500**, such as any tool listed in this application. This embodiment of FIG. **15** may be applied to any of endoscope embodiments described above or below to generate alternative embodiments.

[0104] FIGS. **10A** and **10B** illustrate a method for advancing and retracting the suction shaft **316** in the ear visualization system **300**, according to one embodiment. In FIG. **10A**, the physician is depressing the thumb depress portion **314** of the suction component **310** with her thumb T. This advances the suction shaft **316** through the handle **304** and the tool coupler **307a**, thus advancing the suction shaft **316** along the side of the visualization shaft **306**. Thus, the distal end **317** of the suction shaft **316** would be advanced farther down into the patient's ear. In this configuration, the spring **318** is compressed. In FIG. **10B**, the physician has released her thumb T from the thumb depress portion **314**, allowing the spring **318** to expand and causing the suction shaft **316** to retract proximally through the tool coupler **307a** and the handle **304**. Thus, the physician can easily adjust the position of the distal end **317** of the suction shaft **316** relative to the visualization shaft **306**.

[0105] In various embodiments, the distal end **317** of the suction shaft **316** may be positioned in a number of different locations relative to the distal end of the visualization shaft **306**. When the suction shaft **316** is fully advanced, its distal end **317** may be located at, proximal to or distal to the distal end of the visualization shaft **306**. Similarly, when the suction shaft **316** is fully retracted, its distal end **317** may be located at, proximal to or distal to the distal end of the visualization shaft **306**. For example, in one embodiment, the distal end **317** of the suction shaft **316** may be disposed even with the distal end of the visualization shaft **306** in the fully retracted position and then may be advanced to a position distally beyond the distal end of the visualization shaft **306**. In another embodiment, the distal end **317** of the suction shaft **316** may be disposed more proximally than the distal end of the visualization shaft **306** in the fully retracted position and then may be advanced to a position even with the distal end of the visualization shaft **306**. Any combination of locations is possible, according to various alternative embodiments.

[0106] Referring to FIG. **11**, a physician's hand His shown holding the combination device **300** over an anatomical model. As shown, the handle **304** fits comfortably in the hand H, with the middle finger on the bottom and the index finger on the top. The thumb is positioned on the thumb depress portion **314**, and the visualization component shaft **306** and the suction shaft **316** are extended into the model. During an ear procedure, the physician might rest his or her hand on the patient's head, for support and stability and to prevent arm fatigue. The very light weight of the handle **304** and the device **300** in general make it easy to manipulate and hold.

[0107] Referring now to FIG. **12**, in some embodiments, the ear visualization system **300** may include the ear endoscope **302** and a viewing system **330**. The viewing system may include a video monitor **336**, a console **332** and a cable **334** connecting the two. The console **332** may include a connector **338**, into which a connector **337** on the visualization component **302** inserts. The various parts of the viewing system **330** may be any suitable off-the-shelf or custom components, according to various embodiments. In an alternative embodiment, the console **332** may include a built-in screen, rather than having a separate video monitor **336**, and the endoscope **300** would connect to the console **332**. In various embodiments, the ear endoscope **302** may be provided with the viewing system **330**, with the suction device **310** or as a stand-alone device.

[0108] Referring now to FIGS. **13A-13C**, another embodiment of an ear endoscope **400** is illustrated. In this embodiment the ear endoscope **400** includes a handle **402** with a finger loop **404**, a shaft **406**, two tool coupling shafts **408a**, **408b**, two side suction tube connection ports **410a**,

410b, two suction tube insertion ports **412a**, **412b**, a rear suction tube connection port **414** and a sensor interface cable **416**. In this embodiment, the ear endoscope **400** may be provided as a separate unit and may be used with an add-on suction device, or it may be provided with the suction device. In either case, a suction supply may be connected to the rear suction tube connection port **414**, which is in fluid communication with a suction lumen running through the handle **402** and exiting at the two side suction tube connection ports **410a**, **410b**. One of the two side suction tube connection ports **410a**, **410b** may in turn be connected to a short suction tube, which is connected to a suction shaft that passes through one of the suction tube insertion ports **412a**, **412b** and one of the tool coupling shafts **408a**, **408b**, as will be described further below. Whichever of the two suction tube connection ports **410a**, **410b** is left open may be used by the physician as a suction control, by placing a finger over the open port **410a** or **410b** to apply suction or releasing the finger from the open port **410a** or **410b** to turn off suction.

[0109] The finger loop **404** on the handle **402** may be flexible in some embodiments and rigid in others. In alternative embodiments, the finger loop **404** may have any other suitable shape or size for facilitating gripping the endoscope **400**. The finger loop **404** allows the physician user to hold and operate the ear endoscope **400** with one hand. In fact, the finger loop **404** may allow the physician to hold the handle **402** with one finger (middle finger, for example) and operate other functions of the device **400** with other fingers. For example, the user may pass a middle finger through the finger loop **404**, use the thumb of the same hand to advance the suction tube, and use the index finger of the same hand to control suction by covering and uncovering the open port **410a** or **410b**. Alternatively, the fingers may be placed and used in a different configuration on the handle **402**. As is evident from FIGS. **13A-13C**, the shaft **406** is straight in this embodiment, but it is angled relative to the handle **402**, so that the overall endoscope device **400** is angled, to allow the physician to place the shaft **406** in the ear canal without obstructing a direct viewing path into the ear canal.

[0110] Referring now to FIGS. **14A-14C**, the ear endoscope **400** of FIGS. **13A-13C** is now shown with an optional suction device **420**. FIG. **14A** shows the suction device **420** detached from the ear endoscope **400**. The suction device **420** includes a suction shaft **422**, connected to a thumb depress member **424**, which is connected to a side suction tube **426**. A spring is disposed over a proximal portion of the suction shaft **422**, to provide for automatic retraction of the suction shaft **422** relative to the visualization shaft **406** of the ear endoscope **400**, when the user releases pressure off of the thumb depress member **424**. The suction device **420** may also include a rear suction tube **428**. The suction shaft **422** is passed through either of the two suction tube insertion ports **412a**, **412b** and thus through the corresponding tool coupling shaft **408a**, **408b**. Side suction tube **426** may be connected to either of the two side suction tube connection ports **410a**, **410b**, leaving the opposite side port **410a**, **410b** open for finger control of suction. Additionally, the rear suction tube **428** is attached to the rear suction tube connection port **414**, to supply suction force from a suction supply (such as wall suction, canister suction, etc.-not shown in FIG. **14A**) to the suction device **420**. As explained above, a suction lumen in the handle **402** of the endoscope **400** (not visible in the figures) connects the rear suction tube connection port **414** with the two side suction tube connection ports **410a**, **410b**.

[0111] FIG. **14B** shows all the components of the suction device **420** attached to the ear endoscope. As shown in this figure, the spring **418** resides over the suction shaft **422** and between a bottom surface of the thumb depress member **424** and a top surface of the handle **402**. When the user presses down on the thumb depress member **424**, the suction shaft **422** advances distally along the visualization shaft **406**, and the spring **418** compresses. When the user then releases pressure off of the thumb depress member **424**, the suction shaft **422** automatically retracts proximally, relative to the handle **402** and the visualization shaft **406**.

[0112] FIG. **14C** shows a physician's left hand **H** holding the combined ear endoscope **400** and suction device. As illustrated here, the physician's thumb is positioned on the thumb depress

member **424** and is used to advance the suction shaft **422**. The physician's middle or ring finger may be placed through the finger loop **404** of the handle **402**, and the physician's index finger may be placed over or removed from the open side suction tube connection port **410b**, to control the application of suction through the suction shaft **422**. If the physician prefers to hold the ear endoscope **400** in his right hand, the side suction tube **426** and suction shaft **422** may simply be shifted to the opposite side of the ear endoscope **400**.

[0113] FIG. **15** is a cross-sectional view of a distal/shaft portion of an ear visualization device **500**, according to one embodiment. The portion of the device **500** illustrated in FIG. **15** may be used with any of the embodiments described above or below. The illustrated portion of the device **500** includes an outer shaft **502**, inside of which there is a suction shaft **504** forming a suction lumen **506**. Also located inside the outer shaft **502** are a camera **510** and two sets of light fibers **508a**, **508b**, which are illustrated as rectangular but may be bundled in circular, ovoid or any other suitable shapes. In some embodiments, the inside of the outer shaft **502** may be solid or filled with a material, and the camera **510** and light fibers **508a**, **508b** may reside in lumens formed within the material in the outer shaft **502**. The camera **510** may be located at or near the distal end of the outer shaft **502** and may be any kind of suitable camera, such as but not limited to a complementary metal-oxide-semiconductor (CMOS) or charge-coupled device (CCD) camera. The light source may be any type of suitable light source, as well, including but not limited to light fibers **508a**, **508b**, one or more light emitting diodes (LEDs), etc. Again, this is only one exemplary embodiment of the shaft portion of a device **500**.

[0114] FIGS. **16A** and **16B** illustrate another embodiment of an ear visualization/suction device **600**. As with previous embodiments, device **600** includes a handle **602**, a visualization shaft **606** (or "endoscope shaft," which includes a camera at its distal end), a suction guide shaft **607** on the side of the visualization shaft **606**, a thumb depress portion **624** attached to a suction shaft **622** and a suction tube **626**, and a spring **618** disposed over a portion of the suction shaft **622** to automatically retract the suction shaft **622** when the user releases her thumb from the thumb depress portion **624** (as in FIG. **16B**). In this embodiment, the suction shaft **622** of the ear visualization/suction device **600** further includes a curved distal tip **623**, and the thumb depress portion **624** includes a surface feature **628**. The physician user may apply pressure to the thumb depress portion **624** (as in FIG. **16A**) and use the surface feature **628** (a bump in the present embodiment) to spin the thumb depress portion **624** and thus the suction shaft **622** and the curved distal end **623**. In this way, the suction distal end **623** may be directed in multiple different directions to suction different areas within the ear. The curved distal tip **623** may have different shapes and angles and the surface feature **628** may also have many different shapes and configurations, some of which are described in further detail below. For example, in one alternative embodiment, the extreme end of the curved distal end **623** may be slanted, rather than the blunt/flat end illustrated in FIGS. **16A** and **16B**, which may help facilitated suctioning over surfaces as the distal end **623** is spun around.

[0115] FIG. **17** illustrates another embodiment of an ear visualization/suction device **700**. This embodiment also includes a handle **702**, which in this case is attached at its proximal end to an endoscope cable **730** and a suction tube **732**, both of which pass through the body of the handle **702** to the front/distal end of the handle **702**. Suction tube **732** passes through the body of the handle **702** and terminates at two suction ports **708** (only the left port is visible in FIG. **17**), one on each side of the handle **702**. One suction port **708** is attached to a short suction tube **712**, which in turn attaches to a suction port **710** on a free spin suction member **715** that is disposed over, and spins freely around, the suction shaft **722**. The suction port **710** on the free spin suction member **715** is in fluid communication with the lumen of the suction shaft **722**, so that suction force passes through the port **710** and into the lumen of the shaft **722**. In use, suction travels from the suction tube **732**, through the handle **702**, through the suction port **708**, through the short suction tube **712**, through the suction port **710**, and down the suction shaft **722** to the distal end of the suction shaft **722**. The physician user will cover an opposite-side, open suction port (corresponding to **708**, but on the side

of the handle **702** that is not visible in FIG. **17**) with his finger to apply suction to the suction shaft **722** and will release his finger from the open suction port to allow suction to flow out of the open port and thus stop the flow of suction through the suction shaft **722**.

[0116] As with the previously described embodiment, the ear visualization/suction device **700** includes a visualization shaft **706**, which includes a camera at its distal end for visualizing the ear. It also includes a suction guide shaft **707**, and a suction shaft **722** with a curved distal tip **723**. The device **700** may also include a spring **718** disposed over the proximal portion of the suction shaft **722**, between the top of the handle **702** and the bottom of the free spin suction member **715**. The free spin suction member **715** is disposed over the proximal end portion of the suction shaft **722**, just below the thumb depress member **724**, which again includes a surface feature **728**. The free spin suction member **715** houses two O-rings **714**, one of which is positioned above the hole in the suction tube **722** into which the port **710** leads, and one of which is below the hole. The free spin suction member **715** and O-rings **714** form a liquid sealed chamber that will pass suction from the short suction tube **712**, through the suction port **710** on the free spin suction member **715** and finally into the suction shaft **722**. At the same time, the suction shaft **722** can be rotated, while the free spin suction member **715** stays in one place (does not rotate), thus preventing the short suction tube **712** from twisting around the suction shaft **722**. Again, in this embodiment, when the user spins the thumb depress member **724**, using the surface feature **728**, the free spin suction member **715** does not rotate with the thumb depress member **724** or the suction shaft **722**, but instead remains in the same place, by rotating freely over the suction shaft **722** as the shaft spins. In this way, the suction port **710** does not spin together with the suction shaft **722** but stays in one place. This is important, because if the free spin suction member **715**, along with the suction port **710**, were to spin around with the thumb depress member **724** and the suction shaft **722**, the short suction tube **712** would twist around the suction shaft **722** and make it difficult for the user to spin the suction shaft **722**. It would also prevent free, continuous, 360-degree rotation of the suction shaft **722** and could potentially also result in pulling off of one of the suction ports **708**, **710**, spinning the suction shaft **722** back to its original orientation, or even hampering advancement of the suction shaft **722** in and out of the ear. Advantageously, the free spin suction member **715** allows the user to spin the thumb depress member **724** and the suction shaft **722** continuously, 360 degrees or more, as much as desired, without twisting the short suction tube **712**.

[0117] One additional feature of this embodiment of the ear visualization/suction device **700** is an adjustable finger loop **704** on the handle **702**. In this embodiment, the user may place her finger (for example her middle or index finger) through the looped portion of the finger loop **704** and pull the free end of the finger loop **704** to tighten the looped portion. The looped portion may also be pulled, to make it larger. This is but one example of a finger hold member that may be included on the handle **702** to facilitate holding of the handle **702** using one finger. In one embodiment, the user may hold the device **700** with her middle finger, via the finger loop **704**, and operate the device **700** with other fingers of the same hand (index finger and thumb, for example).

[0118] FIGS. **18A-18D** are perspective views of a portion of an ear visualization/suction device, according to four different embodiments, where each embodiment includes a differently configured thumb depress member for rotating a suction shaft with a curved distal tip. Referring to FIG. **18A**, one embodiment of an ear visualization/suction device **800** includes a generally circular thumb depress member **802** with an elliptical, off-center, raised surface feature **804**. The thumb depress member **802** also includes a slightly concave upper surface for receiving the thumb. Having an off-center surface feature **804** allows the user to rest a finger (typically the thumb) against the surface feature **804** and ergonomically easy spinning of the suction shaft. Also pictured in this and the following embodiments are a handle **801**, a handle suction port **808**, a suction shaft **810**, and a suction shaft suction port **806**.

[0119] In the embodiment of FIG. **18B**, the ear visualization/suction device **810** includes a similarly circular, slightly concave thumb depress member **812**, with a more round shaped (or oval)

surface feature **814**. In the embodiment of FIG. **18C**, the ear visualization/suction device **820** includes an asymmetrical, bean-shaped thumb depress member **822** with a round-shaped (or oval) raised surface feature **824** on a lateral portion of the thumb depress member **822**. In the embodiment of FIG. **18D**, the ear visualization/suction device **830** includes another asymmetrical thumb depress member **832** with a raised, upwardly curving edge at one side. As FIGS. **18A-18D** illustrate, in any given embodiment, the thumb depress member and the surface feature may have any suitable shape, size and configuration.

[0120] Referring now to FIG. **19**, another embodiment of an ear visualization/suction device **900** is illustrated. This embodiment includes many of the features of previous embodiments, which will not be explained again here. In addition, this embodiment of the device **900** includes an adjustable handle **902**, which may make holding the handle **902** more comfortable for a given user. The handle **902** has a distal portion **904**, a proximal portion **906**, and an adjustment member **908** between the two. The adjustment member **908** acts as an axis, to allow the proximal portion **906** to rotate down (and back up as desired), so that the proximal portion **906** can be angled, relative to the distal portion **904**. Thus, the user can direct the tip of device **900** in the direction of the ear canal (typically superiorly-posteriorly), regardless of whether the device **900** is being used in a left ear or a right ear, without changing the orientation of the user's entire hand. In various embodiments, the adjustment member **908** may have specific adjustment angles that the proximal portion **906** clicks into and/or may have a locking feature to allow the user to lock the proximal portion **906** in a desired angle. In alternative embodiments, the adjustment member **908** may be positioned farther distally or farther proximally along the body of the handle.

[0121] Referring now to FIG. **20**, in another embodiment, an ear visualization/suction device **1000** may include a handle **1002** that is malleable or includes a malleable section **1004**. As with the similar embodiment, the malleable section **1004** allows the user to create an angle in the handle **1002**, which may enhance ergonomics and comfort.

[0122] Referring now to FIGS. **21A** and **21B**, one embodiment of an ear tube placement device **1100** is illustrated. In this embodiment, the ear tube placement device **1100** also functions as an ear visualization/suction device. Ear tubes are often placed in tympanic membranes TM (or “ear drums”) that lie between the outer ear (or “ear canal,” labeled in the figures as EC) and the middle ear, primarily in children but also adults with frequent ear infections or other problems involving the middle ear. Although one handle/suction tube configuration is illustrated in FIGS. **21A** and **21B**, alternative embodiments may include any of the handle/suction configurations described in this disclosure. As with previously described embodiments, the device **1100** includes a handle **1102**, a thumb depress portion **1104**, a suction tube **1106**, a visualization shaft **1114**, a suction guide tube **1112**, a suction shaft **1110**, and a spring **1108** positioned over a proximal portion of the suction shaft **1110**. This embodiment further includes a stop member **1120** located near the distal end of the suction shaft **1110** and a sharp distal tip **1116** on the suction shaft **1110**. An ear tube **1118** may be part of the device **1100** or, more likely, the ear tube **1118** may be any currently available or to-be-developed ear tube, which is placed over the distal end **1116** of the suction shaft.

[0123] FIG. **21A** illustrates the ear tube placement device **1100** in a pre-deployment position, with the ear tube **1118** mounted on the suction shaft **1110** and the distal end of the device **1100** being advanced into the ear canal EC. The thumb depress member **1104** is not yet depressed. FIG. **21B** shows the device **1100** in position to deploy the ear tube **1118** in the tympanic membrane TM. In use, the suction shaft **1110** will typically be advanced toward the TM, until the sharp distal tip **1116** of the suction shaft **1110** pushes against the tympanic membrane TM to pierce the membrane. Next, the suction shaft **1110** will be further advanced, to push the ear tube **1118** through the new hole in the tympanic membrane TM, as illustrated in FIG. **21B**. While **1118** is pushed through the hole, the stop member **1120** prevents **1118** from being pushed back by the TM and sliding back over suction shaft **1110**, as the suction shaft **1110** advances through the TM. The stop member **1120** may be any suitable piece or surface feature on (or of) the suction shaft **1110**, such as but not limited to a

washer-like member, a raised ring on the suction shaft **1110**, another tube disposed over the suction shaft **1110** or the like. In general, any piece or feature that will prevent the ear tube **1118** from sliding proximally during deployment will be suitable. After the ear tube **1118** is placed in the tympanic membrane TM, the suction shaft **1110** and/or the entire device **1100** may be retracted/pulled back, to slide the suction shaft **1110** out of the ear tube and the device **1100** out of the ear canal.

[0124] FIG. **22** illustrates a distal end of an alternative embodiment of an ear tube placement device **1200**. In this embodiment, the visualization shaft **1202** includes a camera (not visible), as with previously described embodiments. The suction shaft **1204** is curved and ends in a sharp distal tip **1208**. A stop member **1210** is disposed over the suction tube, and an ear tube **1206** is shown in position. In some embodiments, it may be advantageous to have a curved suction shaft **1204** with the ear tube placement device **1200**, as this will allow it to reach different areas of the TM with slight spins of the suction shaft **1204**.

[0125] Referring now to FIGS. **23-26**, in one embodiment of an ear visualization/suction device **1300**, an optional handle angle adjustment member **1310** may be provided. FIG. **23** shows a portion of the device **1300** without the handle angle adjustment member **1310**, illustrating a handle **1302**, with a finger loop **1306**, and a visualization shaft **1304**. FIG. **24** shows the removable handle angle adjustment member **1310**, which has a straight hand rest portion **1312**, a curved portion **1314** and an attachment portion **1316** that attaches to the handle **1302**. The attachment portion **1316** may be slid over the handle **1302** into place, such that the diameter of the handle **1302** and/or a stop member along the handle **1302** helps position the attachment portion **1316**. In some embodiments, the attachment portion **1316** may be moved along the handle **1302** by the user, to any suitable location on the handle **1302**. In various embodiments, the optional handle angle adjustment member **1310** may have any number of different sizes, shapes and angles. The curved portion **1314**, for example, may have any suitable angle. In some embodiments, the curvature of the curved portion **1314** may be adjustable by the user—for example, it may be malleable. In some embodiments, the device **1300** may be provided with multiple, differently shaped handle angle adjustment members **1310**, each having a different angle, shape and/or size.

[0126] FIG. **25** shows a user holding the portion of the device **1300** without the handle angle adjustment member **1310**. This figure illustrates a first relative angle between the user's hand and the handle **1302**, as well as a first orientation of the visualization shaft **1304** of the device **1300**, relative to the hand. FIG. **26** shows the same user's hand holding the device **1300** with the handle angle adjustment member **1310** attached. As seen here, the handle **1302** is now angled in a more upward direction, relative to the user's hand, and the visualization shaft **1304** of the device **1300** is angled toward the hand more than it was in FIG. **24**. At the same time, however, the overall position of the user's hand is almost exactly the same in FIGS. **25** and **26**. This illustrates that the handle angle adjustment member **1310** (or alternate embodiments thereof) may help a user access and visualize different parts of an ear without needing to assume awkward or uncomfortable hand positions.

[0127] FIGS. **27A** and **27B** are diagrammatic illustrations of one embodiment of a combined ear visualization/suction device **1400**, shown in an orientation as if the device **1400** were being held in a left hand while inserted into a right ear (FIG. **27A**) and a left ear (FIG. **27B**). These two figures are simple representations of the device **1400**, to illustrate how a working area in an ear canal might be limited in some scenarios. For example, FIG. **27B** shows a representation of a left ear canal LEC and an area of interest in the left ear LE. The “area of interest” may refer to an area in the ear and/or one or more structures in the ear, such as but not limited to the tympanic membrane, an area just beyond the tympanic membrane in the middle ear, an area just before the tympanic membrane, or any other suitable area and/or structure in the ear. The device **1400** is being held in the left hand of the user (hand not shown), and the angle of view is represented in the figure by an eye symbol at the distal tip **1412** of the device **1400**. The device **1400** includes a proximal bend **1410**, which

allows for a working area **1420** within the left ear canal LEC. The working area **1420** can be used for passing one or more tools for performing a procedure on the left ear LE area of interest. As FIG. **27B** illustrates, when a user holds the device **1400** in his left hand and accesses the left ear canal LEC with the device **1400**, the working area **1420** is sufficient for passing one or more tools, because it has a relatively large entry opening (the opening into the ear canal from outside the patient).

[0128] In contrast, referring to FIG. **27A**, when the left handed user uses the device **1400** to access a right ear canal REC to view an area of interest in the right ear RE, this embodiment of the device **1400** may cut off the entry opening of a working area **1414**, which may make it more difficult for the user to insert working tools into the ear canal REC. Although the overall size of the working area **1414** in FIG. **27A** is approximately the same as the size of the working area **1420** in FIG. **27B**, the configuration of the working area **1414** for a left handed user in a right ear canal REC may be problematic for passage of tools. This may require the user to position his/her hand in an awkward position or insert tools through the ear canal REC in an awkward way. The same problem may occur if a person holds the device **1400** in the right hand and uses the device **1400** in the left ear canal LEC.

[0129] FIGS. **28A** and **28B** are diagrammatic illustrations of an alternative embodiment of a combined ear visualization/suction device **1500**, shown in an orientation as if the device **1500** were being held in a left hand while inserted into a right ear (FIG. **28A**) and a left ear (FIG. **28B**). In this embodiment, the device **1500** includes a proximal bend **1510** and a distal bend **1520**, closer to the distal end **1522** of the device **1500**. Thus, when held in the user's left hand and inserted into the right ear canal REC, the resulting working area **1524** has a sufficiently large opening and overall width to allow for tool passage. The distal bend **1520** may be formed in any of a number of ways and may be either fixed or adjustable in various embodiments. For example, in some embodiments, a distal portion of the device **1500** may be malleable, so the user can form and adjust the distal bend **1520**. In other embodiments, the distal bend **1520** or a distal portion of the device **1500** may be rotatable. Alternatively, the view of the camera at the distal end **1522** of the device **1500** may be rotated or otherwise adjusted, for example by electronic adjustment, to allow for the desired viewing angle.

[0130] FIG. **28B** shows the device **1500** with the distal bend **1520** adjusted to have a different angle, designed to access the left ear canal LEC and view the left ear area of interest LE. In this scenario, the working area **1530** is also sufficiently wide to allow for tool passage. Various alternative embodiments may have any number, location and angle of bends. In one embodiment, a left-handed device and a right-handed device may be provided.

[0131] Referring now to FIGS. **29A** and **29B**, an alternative embodiment of an ear visualization and suction device **1600** is illustrated. In various embodiments, the suction tube of an ear visualization/suction device (the side tube through which the suction device passes) may have an inner diameter ranging from about 0.6 mm to about 2.5 mm, with some embodiments having an inner diameter of about 1.1 mm or less. The visualization shaft (or visualization shaft) may have an outer diameter of about 2.5 mm or less or less than about 5 mm. These dimensions may differ in some embodiments. In the embodiments described above, the suction shaft (i.e., the part of the device that transmits suction force to the distal end of the device) passes through the suction tube and slides up and down through the tube as it is advanced and retracted. The size of the outer diameter of the suction shaft is thus limited by the size of the inner diameter of the suction tube. In various embodiments, for example, the suction shaft may have an outer diameter ranging from about 0.6 mm to about 2.5 mm, depending on the inner diameter of the suction tube. In some embodiments, the suction shaft of the ear visualization/suction device may have an outer diameter measuring about 3 French (i.e., about 1 mm). For some ear surgery procedures, this diameter works well. In other cases, however, it might be advantageous to have a larger suction shaft diameter, to provide more suction force and/or prevent clogging of the suction device. For example, in some

embodiments it might be advantageous to have a suction shaft with an outer diameter of about 5 French (i.e., about 1.67 mm). Any number of suitable sizes of suction shafts may be used in a given embodiment, again ranging from as small as about 0.6 mm outer diameter to as large as about 2.5 mm outer diameter or more.

[0132] If the suction shaft must pass through the suction tube, then a larger suction shaft will require a larger suction tube. Therefore, one way to provide ear visualization/suction devices with different suction diameters is to provide separate devices with suction tubes and suction shafts having different diameters, allowing the user to choose between devices for a given procedure or even to swap out devices during a procedure. Another way to allow for suction size variation is to provide one device with multiple, differently sized suction shaft components that may be swapped out on the same device. The ear visualization and suction device **1600**, shown in FIGS. **29A** and **29B** is one embodiment of an adjustable suction size device.

[0133] In the embodiment of FIGS. **29A** and **29B**, the device **1600** includes a handle **1602**, a visualization shaft **1606** (or “visualization shaft”), two suction tubes **1604** extending along either side of a proximal portion of the visualization shaft **1606**, suction ports **1608** on either side of the handle **1602**, a suction shaft **1612**, suction tubing **1610**, a suction push rod **1616**, and a thumb depress **1614** attached to the proximal end of the suction push rod **1616**. In FIG. **29A**, solid-tipped arrows show the path of suction into the distal end of the suction shaft **1612**, into the suction tubing **1610**, which is attached to the handle **1602** via one of the ports **1608**, and then into and through the handle **1602**. In some embodiments, for example, the handle **1602** houses additional suction tubing (shown in later figures) that connects to further tubing to couple with a source of suction. In operation, the user presses the thumb depress **1614** to advance the suction push rod **1616**, which extends through one of the two suction tubes **1604**. The suction push rod **1616** advances the suction shaft **1612**. When suction is activated, as illustrated in FIG. **29A**, suction passes through the suction shaft **1612**, the suction tubing **1610**, and the handle **1602**. The suction push rod **1616** serves merely to advance and retract the suction shaft **1612**. (Retraction occurs automatically via a spring positioned over the proximal portion of the suction push rod **1616**, between the top of the handle **1602** and the bottom of the thumb depress **1614**, as described in further detail above, in regard to other embodiments.)

[0134] FIG. **29C** shows the suction component **1618** of the device **1600**. When coupled with the handle **1602**, the suction push rod **1616** extends from a proximal end attached to the thumb depress **1614** to a distal end attached to the suction shaft **1612** via a junction **1617**. The suction shaft **1612** includes a proximal curve, and its proximal end fits within the suction tubing **1610**. As is evident from FIG. **29C**, the outer diameter of the suction push rod **1616** is smaller than the outer diameter of the suction shaft **1612**. For example, the suction shaft **1612** may have a 5 French outer diameter in one embodiment. In alternative embodiments, the suction shaft **1612** may have any suitable diameter. As long as the suction tube push rod **1616** is sized to fit through the suction tube **1604**, the suction shaft **1612** may have any desired inner and outer diameter, according to different embodiments. In this way, the suction component **1618** and various sizes thereof may allow for the same ear visualization/suction device **1600** to be adjusted to provide multiple different sizes of suction shafts **1612**. The thumb depress **1614** may be easily removable from the proximal end of the suction push rod **1616**, for example via threaded connection or friction fit. In this way, the device **1600** facilitates changing of the suction component **1618** for a different suction component with a differently sized suction shaft **1612**.

[0135] FIG. **30** shows the suction component **1618** on the right and an alternative suction component **1622** on the left. The suction component **1618** was already described in reference to FIG. **29C**, but it is shown here with an alternative thumb depress **1620** and with the spring **1621**. The alternative suction component **1622** includes a suction shaft **1624**, a second alternative thumb depress **1628** with a suction port **1627**, suction tubing **1626** and a spring **1629**. The suction shaft **1624** of the alternative suction component **1622** extends through the suction tube **1604** of the

device **1600**, as described in relation to other embodiments herein. In one embodiment, the suction shaft **1624** may have an outer diameter of about 3 French, while the suction shaft **1612** may have an outer diameter of about 5 French. In one embodiment, the device **1600** may be provided with both suction components **1618**, **1622**, to allow a user to choose between the two for a procedure or to swap between the two during a procedure. One method for swapping out the suction components **1618**, **1622** is to remove the thumb depress **1620**, **1628** (along with the spring **1621**, **1629**) and slide either the suction push rod **1616** or the suction shaft **1624** distally out of the suction tube **1604**. The other suction component **1618**, **1622** may then be loaded into the device **1600** by reversing the process.

[0136] Referring now to FIGS. **31A-31C**, the ear visualization and suction device **1600** is illustrated with the alternative suction component **1622** attached. FIG. **31A** shows a proximal suction tube **1630** extending through the handle **1602**. FIGS. **31B** and **31C** are top cross-sectional views, with the handle **1602** not illustrated. As illustrated by solid-tipped arrows in FIG. **31B**, in this configuration, suction travels up the suction shaft **1624**, through the thumb depress **1628**, through the suction tubing **1626**, and then into and through the proximal suction tubing **1630** in the handle **1602**. FIG. **31C** shows how the suction shaft **1624** extends through the suction tube **1604** and attaches to a distal shaft portion **1632** of the thumb depress **1628**. The distal shaft portion **1632** travels back and forth within a bore **1634** of the handle, with proximal movement assisted by the spring **1629**.

[0137] The embodiments and features illustrated in FIGS. **29A-31C** represent merely one example of an ear visualization and suction device **1600** that provides two different suction shaft diameters. Alternative embodiments may achieve the same objective using different techniques and features. For example, in one embodiment, an ear visualization and suction device may have a larger diameter suction tube, for example allowing for passage of a 5 French (outer diameter) suction shaft. That device may also be provided with an alternative suction component having a suction shaft with a smaller outer diameter, such as 3 French. The smaller suction shaft may have a wider proximal portion that fits in the suction tube, to prevent the shaft from wiggling or jostling within the suction tube. Thus, any number of different embodiments and sizes of suction shafts may be provided as part of an ear visualization/suction device.

[0138] Referring now to FIGS. **32A-32D**, in one embodiment, an ear visualization and suction device **1700** may also include an irrigation component. FIG. **32A** shows an assembled side view of the device **1700**, which includes a handle **1702**, with a finger hold **1704**, an electric cable **1706**, a proximal suction tube **1708**, an irrigation tube **1710**, a distal suction tube **1714**, a thumb depress **1712** and a suction shaft **1716**. (The visualization shaft is hidden behind the suction shaft **1716** in this view.) The electric cable **1706** may contain, for example, image sensor wires and LED power lines. The device **1700** includes other parts and features described in relation to other embodiments above, but these will not be described again here. In this embodiment, the suction shaft **1716**, extends through a first bore in the top of the handle **1702**, and an irrigation shaft (not visible in FIG. **32A**) extends through a second bore. Irrigation fluid, such as saline solution, is passed through the irrigation tube **1710** and irrigation shaft so that it flows down the side of the visualization shaft into the patient's ear. Suction from the suction shaft **1716** may be used to evacuate the irrigation fluid.

[0139] FIG. **32B** shows the ear visualization and suction device **1700** with the irrigation shaft **1722** removed from the handle. As mentioned above, to use the irrigation portion of the device **1700**, the irrigation shaft **1722** is passed down into one of two bores **1720** on the top of the handle **1702**, and thus into one of the two suction tubes **1718** that extend along either side of a proximal portion of the visualization shaft **1717**. Either bore **1720** may be used, and the suction shaft **1716** (not shown in FIG. **32B**) is inserted into the other bore **1720**. As evident from FIG. **32B**, the irrigation shaft **1722** is shorter than the visualization shaft **1717**, and it may also be shorter than the length of the two suction tubes **1718**. Thus, when irrigation fluid passes out of the distal end of the irrigation

shaft **1722**, it passes along the side of the visualization shaft **1717** and into the ear. In alternative embodiments, the irrigation shaft **1722** may be longer or shorter than the one shown. Also visible in FIG. **32B** is one of two handle suction ports **1724**, on either side of the handle **1702**, configured for coupling with the suction tube **1714** (FIG. **32A**).

[0140] FIG. **32C** is the same side view as in FIG. **32B**, but with the irrigation shaft inserted into one of the bores **1720**. FIG. **32D** is a top view of the same configuration of the ear visualization and suction device **1700**.

[0141] Referring now to FIGS. **33-36**, an alternative example of an ear endoscope device **1800** is illustrated. In this embodiment, as shown in FIG. **33**, the ear endoscope device **1800** includes a handle **1802**, a shaft **1804** with a distal tip **1806**. As illustrated in FIG. **34**, the distal tip **1806** is angled, rather than having an end that is perpendicular to the longitudinal axis of the shaft **1804**. For example, the distal tip **1806** may be angled at an angle of between about 5 degrees and about 60 degrees, or between about 10 degrees and about 45 degrees, or between about 15 degrees and about 30 degrees in various embodiments. In the embodiment shown in FIG. **34**, the distal tip **1806** has an angle of approximately 20 degrees. As illustrated in FIG. **35**, the proximal end of the shaft **1804** is attached to a rotation mechanism **1810**, which connects the shaft **1804** to the handle **1802** in such a way that the shaft **1804** is rotatable about its longitudinal axis.

[0142] In the illustrated example of the ear endoscope device **1800**, the shaft **1804** can rotate relative to the handle **1802**, thus rotating a camera **1809** at the distal tip **1806** of the device **1800** more than 360 degrees in either direction. This allows the user to view the ear (or other structure if used in another part of the body) from many different directions. In one embodiment, as illustrated in FIG. **33**, the ear endoscope device **1800** includes a rotation cuff **1803** and a rotation indicator **1805**. With this embodiment, the user can hold the rotation cuff **1803** and rotate the shaft **1804** in either direction, with the rotation indicator **1805** pointing in the opposite direction of the angled end of the distal tip **1806** of the shaft **1804**. To rotate the shaft **1804**, the user the same hand that is holding the ear endoscope device **1800** or can use the other hand.

[0143] Referring to FIG. **35**, in one embodiment, the rotation mechanism **1810** includes a connecting ring **1816** attached directly to the shaft **1804**, a connector **1818** connected directly to the handle **1802**, and a limiter ring **1814** that is free to rotate around the endoscope shaft and has two limiter ring rotation limiters **1830**. The connector **1818** includes two graspers **1832** which latch onto connecting ring **1816** to connect shaft **1804** to handle **1802** and allow them to rotate with respect to each other. one of the graspers has a limiter bump **1834** that interacts with the limiter ring **1814**. The shaft **1804** includes two rotation limiters **1840** that move with it, indicated in FIG. **36**, which is a partial phantom view of the rotation mechanism of FIG. **35**. When the user rotates the rotation cuff **1803**, a rotation limiter on the shaft **1804** rotates with it, slides under limiter bump without engaging it, and eventually couples with a first limiter ring rotation limiter of limiter ring **1814**. The shaft **1804** and limiter ring **1814** then start rotating together until the second limiter ring rotation limiter of limiter ring **1814** reaches the other side of limiter bump, and cannot rotate further. At this point, rotation is stopped, and the user cannot rotate the shaft **1804** further relative to the handle **1802**. With this rotation mechanism **1810**, the shaft **1804** can rotate more than 360 degrees and in some cases as much as almost 720 degrees. When the shaft **1804** rotates, the wires passing from the camera **1809** and LED **1808** through the shaft **1804** will twist, but the rotation mechanism **1810** prevents them from twisting too much and tearing the wires. The wires for the camera **1809** and LED **1808** pass through a gap **1850** formed by a proximal end **1836** of shaft **1804**, over one side of the connector **1818**, and through the handle **1802**.

[0144] In another embodiment, the distal tip **1806** can include an image guidance sensor coupled thereto so that an image guidance computed tomography (CT) image or anatomical marker can be overlaid on the image received from the image sensor. Examples of such image guidance sensors include, without limitation: motion sensors (e.g., accelerometers and gyroscopes); and position detectors (e.g., ultrasonic or electromagnetic). In yet other alternatives, an image guidance sensor

can be coupled to other structures, such as the suction tip so that, when the suction is advanced and rotated, it acts like an image guidance probe, similar to having the image guidance sensor on the tip of the camera. In another example, an image guidance sensor is coupled to a tip of another tool (e.g., guidewire) so that it can be passed through one or more tool ports, alongside the main shaft of the endoscope (the visualization shaft). Many other configurations are possible.

[0145] The example of a rotating shaft ear endoscope device **1800** illustrated in FIGS. **33-36** is only one example. Many other embodiments and variations of an endoscope device with a rotating shaft and angled distal end/camera angle are contemplated within the scope of this disclosure. For example, in an alternative embodiment, an ear endoscope device may include a rotating shaft and one or more suction devices on either side of the shaft. In some cases, the suction device(s) may rotate around their own axes and/or around the endoscope shaft. In other embodiments, the rotation mechanism may be different. In other embodiments, any suitable camera/distal tip angle may be used. Some embodiments may include a swing prism endoscope to allow the camera angle to be changed by the user. These and other features may be combined with any of the embodiments described above without departing from the scope of the disclosure.

[0146] Referring now to FIG. **37**, an alternative example of an ear endoscope device **1900** is illustrated. In this embodiment, the ear endoscope device **1900** includes all the component parts and features of the embodiment shown in FIGS. **33-36**, so those parts and features will not be described again here. The added feature in the embodiment of FIG. **37** is a rotation dial **1902** on the handle **1802**. The rotation dial **1902** serves the same purpose as the rotation cuff **1803**—i.e., the ability to rotate the shaft **1804** about the shaft's longitudinal axis. But the rotation dial **1902** offer the physician an additional way to rotate the shaft **1804**. The rotation dial **1902** may be conveniently rotated, for example, with the physician's thumb while holding the handle. The rotation cuff **1803**, on the other hand, may be more easily turned using a finger or a thumb and finger. In some cases, using the rotation dial **1902** to rotate the shaft **1804** might be easier to do while holding the handle **1802** with the same hand. In another alternative embodiment (not pictured), an ear endoscope device may include only a rotation dial and not a rotation cuff.

[0147] The above description of embodiments and features of various devices and methods is believed to be complete. The embodiments are meant to exemplary in nature, however, and not exhaustive. Thus, their description should not be interpreted as limiting the scope of the invention.

Claims

1. A device for visualizing a surgical procedure in an ear, the device comprising: a handle; a visualization shaft extending from the handle, the visualization shaft comprising a distal end disposed at an angle that is not perpendicular to a longitudinal axis of the visualization shaft; a rotation mechanism connecting the visualization shaft to the handle, wherein the rotation mechanism is configured to allow the visualization shaft to rotate around the longitudinal axis relative to the handle; an imaging sensor at the distal end of the visualization shaft; and a light source at the distal end of the visualization shaft, wherein the imaging sensor and the light source are positioned at the angle of the distal end relative to the longitudinal axis of the visualization shaft.
2. The device of claim 1, further comprising: a first suction tube extending along a proximal portion of a first side of the visualization shaft; a second suction tube extending along a proximal portion of a second side of the visualization shaft; and a first suction component, comprising: a first suction shaft; a first thumb depress member coupled with a proximal end of the first suction shaft; a first spring disposed over a proximal portion of the first suction shaft, such that when the first suction shaft is advanced and then released, the first suction shaft retracts automatically; and a first suction tubing for connecting at least one of the first suction shaft or the first thumb depress member with the handle.

3. The device of claim 2, wherein the first suction component further comprises a suction push rod attached to the proximal portion of the first suction shaft, wherein the suction push rod has an outer diameter sized to fit through an inner diameter of the first suction tube and the second suction tube, and wherein the first thumb depress member is attached to a proximal end of the suction push rod.
 4. The device of claim 2, further comprising an irrigation component, comprising: an irrigation shaft sized to fit through the first suction tube or the second suction tube; and an irrigation tubing attached at one end to the irrigation shaft.
 5. The device of claim 2, wherein the first suction shaft has an outer diameter of 3 French or less.
 6. The device of claim 1, wherein the angle is between 10 degrees and 45 degrees.
 7. The device of claim 1, wherein the angle is 20 degrees.
 8. The device of claim 1, wherein the handle comprises a finger loop configured to allow the device to be held by a single finger of a user.
 9. The device of claim 1, wherein the rotation mechanism comprises: a connecting ring attached directly to the visualization shaft; a connector connected directly to the handle; and a limiter ring that rotates around the visualization shaft and includes at least one rotation limiter.
 10. The device of claim 1, further comprising: a rotation cuff attached to an outer surface of the visualization shaft at or near the handle; and a rotation indicator attached to the rotation cuff.
 11. The device of claim 1, wherein the rotation mechanism comprises a rotation dial on the handle.
 12. The device of claim 1, further comprising an image guidance sensor at the distal end of the visualization shaft.
 13. A method for visualizing a surgical procedure in an ear, the method comprising: advancing a distal end of an ear visualization device into the ear; visualizing a first portion of the ear with an angled camera at the distal end of a visualization shaft of the ear visualization device; rotating the visualization shaft about a longitudinal axis of the visualization shaft and relative to a handle of the ear visualization device; and visualizing a second portion of the ear with the angled camera.
 14. The method of claim 13, wherein an angle of the angled camera, relative to a plane perpendicular to the longitudinal axis of the visualization shaft, is between 10 degrees and 45 degrees.
 15. The method of claim 13, wherein an angle of the angled camera, relative to a plane perpendicular to the longitudinal axis of the visualization shaft, is 20 degrees.
 16. The method of claim 13, further comprising applying suction in the ear with a suction member of the ear visualization device.
 17. The method of claim 16, further comprising applying irrigation in the ear with an irrigation member of the ear visualization device.
 18. The method of claim 13, wherein rotating the visualization shaft comprises rotating the visualization shaft with a hand that is holding the handle of the ear visualization device, using at least one of a rotation cuff adjacent the handle or a rotation dial on the handle.
 19. The method of claim 13, further comprising monitoring rotation of the visualization shaft and an angle of the angled camera with a rotation indicator coupled with the visualization shaft.
 20. The method of claim 13, further comprising coupling an image guidance sensor at the distal end of the ear visualization device.
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